

What is Flexbox?

- The Flexbox means flexible box layout.
- Flexbox is 1D layout.
- The flexbox is a CSS layout model used to:
 1. Align Items
 2. Provide Spacing
 3. Create Responsive Layouts
 4. Arrange elements in rows or columns easily
- It solves many problems of older layouts like:
 1. inline-block
 2. positioning

Main Concepts in Flexbox:

1. Flex Container: Parent Element
Syntax: `display: flex;`
2. Flex Items: Children inside the flex container.

Flex Axis:

- There are two axis for flexbox:
 1. Main Axis: Direction in which items move.
 2. Cross Axis: Opposite direction

Flex Parent Properties:

1. `display: flex;`
- Converts normal container into flex container.

2. justify-content: start | center | end | space-around | space-evenly | space-between

- Controls alignment on Main Axis.

3. align-items: stretch (default) | start | center | end

-> Controls alignment on Cross Axis.

4. flex-direction: row (default) | column | row-reverse | column-reverse

-> Provide direction for items.

Very Very Important:

flex-direction: row:

justify-content -> Main Axis (Horizontal)

align-items -> Cross Axis (vertical)

flex-direction: column;

justify-content -> Main Axis (Vertical)

align-items -> Cross Axis (Horizontal)

5. flex-wrap: nowrap (default) | wrap | wrap-reverse

-> Controls whether items move to next line.

-> Controls wrapping of items.

6. flex-flow: flex-direction flex-wrap;

-> Shortcut property for : direction and wrap.

7. align-content: start | center | end | space-around | space-evenly | space-between

-> Controls spacing between multiple rows.

-> Works only when:

flex-wrap: wrap

Multiple rows exist

8. gap: row-gap column-gap;

-> Adds spacing between flex items.

Flex Child Properties:

1. order: It is used to change the item order.

2. flex-grow: Defines how much item grows.

3. flex-shrink: Defines shrinking behavior.

4. flex-basis: Defines initial size before growing/shrinking.

5. flex: flex-grow flex-shrink flex-basis

6. align-self: Applies alignment to individual item.

Important Interview Questions:

Q.1. Difference Between justify-content and align-items?

Q.2. Difference Between align-items and align-content?

Q.3. What is the Default Value of flex-direction?

Q.4. What is the Difference Between flex-grow, flex-shrink, and flex-basis?

Q.5. What is flex: 1?